Final Project: Report

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Introduction

The use of statistical methods to uncover hidden insights about baseball continues to surprise us every day. Especially given the importance of the robust analytical insights shown in this research project. Through the implementation of the IL placement logistic regressions, we not only managed to incorporate a unique aspect of the baseball game that adds value to an organization from top to bottom, but also closely understand the behavioral trends of baseball injuries directly linked to pitchers who are indeed the driving force of a team's increasing probability of winning a baseball game.

In our goals of understanding if and how pitching metrics predicted IL placement, and familiarizing ourselves with detrimental factors affecting pitchers in today's game, and how these factors can be mitigated to improve stamina and overall longevity, we managed to demonstrate the impact that proving our points through the backing of properly established statistical procedures has in the immediate developmental effect of a baseball team.

In the game of baseball, players are the primary asset, and the game revolves around them. Our aim in this analytical research paper was to identify all avenues for teams to continue putting proper strategies in place to survive the ups/downs of a long baseball season, which is usually marred by injuries, especially those related to pitchers, who give their maximum effort on the baseball diamond. The evolution of baseball games, as noted in the previous draft, is abysmal, as we continue to see characteristics that have become key components of pitchers in today's era that were not as valued 30-40 years ago. As diverse baseball eras have developed with different pitching philosophies, workload strategies, and the implementation of controversial technological platforms like Driveline, changing the game, leading to an evident spike in injuries. Something that is indeed contradictory, considering that we now have greater medical and technological advancements that can be used to develop better treatments and mitigate the risk of injury. However, this comes down to a straightforward question: what are teams' executives, coaches, and players doing differently now compared to decades ago?

The implementation of the Pitch Clock and the ABS system have a highly significant impact on the industry's marketability by reducing game time and improving umpire calls. Still, it also increases the possibility of a pitcher's IL placement. However, we also know, based on MLB's league reports, that the rise in pitcher injuries is driven by the chase for more velocity, nastier stuff (via pitch design), and max-effort pitching (Adler, 2024). Hence, it was essential for us to take some time to unravel additional hidden insights, trends, and statistical significance of these inferences through the effective incorporation of the IL placement dataset containing data from 2015 to 2017, to effectively determine and project how these factors lead to the detrimental impact on a pitcher's health.

The robust structure built throughout this project's dataset combination (Lahman Pitching, Savant – Statcast level, and IL Data) will guide teams in their risk management and projection planning, enabling them to make effective roster management decisions to protect well-established pitchers and other highly coveted prospects and complementing the results of the initial analysis, which suggested that detrimental factors also affecting a pitcher's longevity in today's game may include elevated walk rates and potentially excessive workloads. At the same time, strikeout ability and controlled velocity appear to be beneficial; combined with previously proven statistical procedures that raise the significance of our analysis in many ways. Knowing that while velocity remains a performance asset, balancing it with control and workload management is a critical consideration in the data complementation of IL placement.

Data Preparation and Integration

The combination of this dataset was performed using the Savant API system, which delivered a CSV, created with specific characteristics that included all relevant Statcast pitching metrics. Then, using R Studio, this CSV was effectively combined with the baseball R built-in library (Lahman), which provides a Pitching dataset table with statistics from 1871 to date, as indicated in the proposal/dataset selection deliverable. Using Normalizer helpers to standardize the names, both datasets were combined by matching names in common and removing all names not included in the Savant dataset. We also filtered the Lahman Pitching dataset to the intended range starting in 2015. Since this version of the library is limited to 2023, an end date was not required. We also began by creating interesting per 9 innings metrics, which were further explored through the EDA analysis. However, one of the most critical aspects of efficiently laying the groundwork for this data was to check for data duplicates when creating the lahman_pitcher_totals data frame, which already included a cumulative statistical compilation of qualified pitchers.

At this moment, it was key to open the CSV separately and manually search for all duplicates to determine if they were present in the Savant dataset. To optimize the identification process of duplicate pitchers with the same names, we accessed TruMedia, which provides baseball teams with robust tools for data analytics, visualizations, and video scouting. Here, we could easily identify the players with their MLB IDs and remove those present in the Lahman Pitching library but not in Savant. Upon manually proceeding with this much-needed adjustment, we then loaded the Savant dataset, only including relevant and interesting variables and by immediately standardizing the names to make them match those in Lahman Pitching dataset. Upon the completion of the data preparation stage, which was thoroughly explained, we proceeded to integrate the IL data, for which it was deemed necessary to create various data subsets to properly prepare the data to be merged and adequately clean the attribute naming structure, eliminating invalid and erroneous data and building join key helpers to integrate data across frames efficiently. Once this initial cleansing stage was demonstrated to be ready for further debugging, we produced an additional cleaning subset, limiting the data to the relevant years (2015-2017), based on MLB's publicly available dataset regarding IL placement. Even though we knew this data included position players, we didn't focus on eliminating them, as they would be limited when merging the IL dataset with the already established pitcher_df containing pitchers with at least 2000 pitches obtained via the Savant dataset incorporation.

Before merging this IL placement data with what we have already constructed, it was essential to add relevant variables to it, built from its already included original data. This is why, at this point, we created a final subset to include only the relevant data for merging considerations. Here, we used join_key_fl as the standardized name variable and included an injury_count variable to reflect the amount of injuries this given pitcher had in this three seasons timeframe; total_dl_days which summarized the amount of days this given pitcher spent on the IL list; last_injury_dt to showcase the last date of this player's injury placement and injury_types to note the types of injuries the player has surpassed. Upon completion of this stage, it was necessary to once again check dupes in our data frame, and it was confirmed that no duplicates were part of this IL data, which enabled it to be ready for merging purposes.

To properly merge both data frames, we mutated the names in pitcher_df to standardize its naming structure and make it simulate the same structure as that of the IL names, which displayed 'First Last names' instead of 'Last, First' as the pitcher_df was initially structured. Then, through a left join, we merged the data. Instead of including injury types and the last date of injury, we included only the number of injuries, the total days spent on the injured list, and the injury occurrence, which was coded as a binary variable for further use in the logistic regression analysis. Ultimately, after dropping the naming helpers and producing a final clean data structure, an optimal sample-size dataset was constructed, yielding a robust dataset of 826 observations with 48 variables, consisting of each qualified pitcher's statistical compilation across these nine seasons, and setting the ground to further analytical procedures.

Methods

Through the combination of the three datasets, we maximized each dataset's potential by extracting essential metrics from each and further using them to complement each other in further analytical procedures. Restricting the data to the Statcast era (2015-2023) was an effective measure to ensure stability of the overall data analysis and avoid further issues of working with missing data that would have minimized the robustness of the report. The restriction of this data also guaranteed the availability of modern pitching metrics and enabled aggregating player performance metrics at a career level to showcase the performance of long-term pitching outcomes.

For every pitcher who appeared from 2015 to 2023, individual season statistics from the Lahman Pitching dataset were grouped and summed across seasons. This included career innings pitched, career earned run average, career strikeouts per nine innings, career walks per nine innings, career average homeruns allowed per nine innings, and career walks plus hits per nine innings. This career-level aggregation enabled merging with the Statcast data (velocity, spin rate, etc.), providing a complete dataset that merges every metric for every pitcher. This dataset included one record for each pitcher, enabling the analysis of relationships between performance metrics and overall results, such as career ERA and total innings pitched, which will be further explored later in the Analysis section.

This analysis employs three methods. An exploratory data analysis was conducted to gain a deeper understanding of the dataset, including its distribution and relationships among the key variables. This included summary statistics, a correlation matrix, and different visualizations as shown in the Initial Analysis Report. Two multivariable linear regression models were also developed: one to model career ERA (a continuous variable) and the other to model innings pitched as a continuous variable. These models allowed the evaluation of both performance and longevity outcomes using the same set of predictors. Two logistic regression models were also developed to predict injury occurrence using workload and performance predictors. This binary model predicts the probability of a pitcher sustaining an injury by analyzing workload intensity and pitching characteristics. Additional diagnostics, such as multicollinearity tests, confusion matrices, ROC Curve, and AUC, were used to ensure model validity and interpretability.

Analysis

Exploratory Data Analysis (EDA)

An exploratory data analysis was conducted to get more insight into the data. As mentioned previously, the pitching_df dataset (combined dataset) contained over 826 observations with 48 variables. A summary statistic of key variables such as ERA career, SO9 career (strikeouts per nine innings), BB9 career (walks per nine innings), WHIP career (walks plus hits per nine innings), velocity, spin rate, and total pitches were analyzed (see Figure 1).

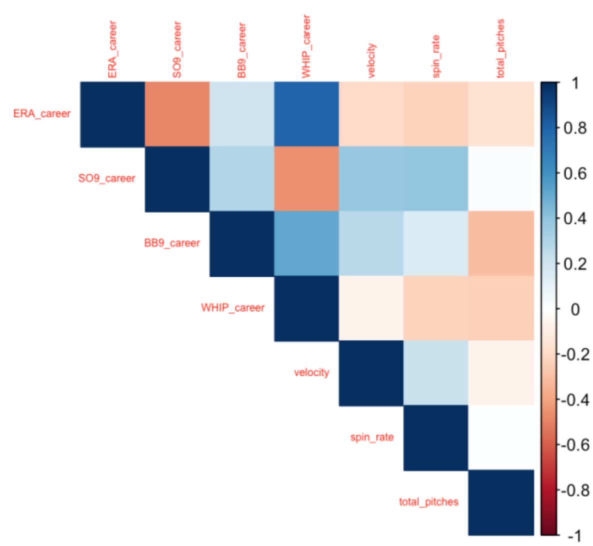
Figure 1

	ERA_career	SO9_career	BB9_career	WHIP_career	velocity	spin_rate	total_pitches
Min.	1.670	3.844	1.075	0.8923	75.00	1649	2001
1st Qu.	3.638	7.366	2.714	1.2229	87.22	2097	3006
Median	4.185	8.479	3.258	1.3165	89.00	2219	4604
Mean	4.194	8.549	3.316	1.3140	88.90	2223	6266
3rd Qu.	4.656	9.630	3.819	1.4026	90.78	2337	7798
Max.	7.000	14.744	6.021	1.7791	97.00	2778	25820

This table revealed interesting information about key pitching metrics in this dataset, such as the average career ERA being around 4.19, with the lowest average being 1.67. The average velocity seemed to be roughly 88.90 mph, and the total pitches averaged by pitchers seemed to be 6,266 pitches, which doesn't seem like a whole lot. Still, we must consider that these are career totals, and some pitchers have probably been in the league longer than others since the data is from the seasons 2015 to 2023.

Using the same key variables as in the descriptive summary statistics table, a correlation matrix and correlation heatmap were produced, as these variables are relevant to performance and pitching metrics. The correlation matrix revealed some meaningful relationships (see Figure 2).

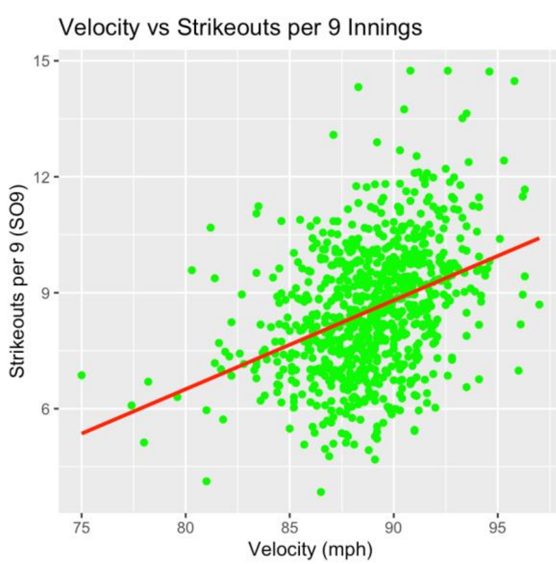
Figure 2



It seems that ERA had a strong positive correlation with WHIP (r ≈ 0.81), which is expected, suggesting that pitchers who allow more baserunners also allow more runs. Strikeout rate (SO9) had a moderate negative correlation with ERA career (r ≈ -0.49), indicating that more dominating pitchers with more strikeouts generally have lower ERAs. SO9 also had a mild positive correlation with both velocity (r ≈ 0.38) and spin rate (r ≈ 0.40), which also suggests that pitchers with greater velocity and spin rates tend to produce more strikeouts. Walk rate (BB9) was moderately correlated with WHIP (r ≈ 0.52) and had a weak positive correlation with ERA (r ≈ 0.21), highlighting the influence of control on performance metrics. Overall, these correlations back the theoretical assumption that higher "stuff" metrics lead to better results.

A scatterplot was also created to look at the relationship between velocity and SO9 (see Figure 3). The correlation plot showed a moderate positive correlation between the two, and the scatterplot supported this finding. The scatterplot showed a positive linear relationship between the two. As velocity increases, so does strikeout rate, indicating that pitchers who throw faster and harder tend to strike out more batters. The linear trend line confirms a moderate positive correlation.

Figure 3



Linear Regression Models

To quantitatively assess the relationships identified in the EDA, two multivariable linear regression models were developed. These models aim to predict key outcomes, pitcher effectiveness (Career ERA), and durability (Total Innings Pitched), using a set of core performance and "stuff" metrics.

Model 1: Predicting Career ERA

The first model (lm_era) was built to predict a pitcher's career Earned Run Average (ERA_career) based on their career strikeouts per 9 innings (SO9_career), walks per 9 innings (BB9_career), average pitch velocity, and spin_rate (see Figure 4).

Figure 4

```
Call:
lm(formula = ERA_career ~ SO9_career + BB9_career + velocity +
    spin_rate, data = pitcher_df)

Residuals:
     Min       1Q   Median       3Q      Max
-2.06657 -0.39430 -0.02311  0.38127  2.40404

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  7.6022032  0.7402143  10.270   <2e-16 ***
SO9_career  -0.2603345  0.0151033 -17.237   <2e-16 ***
BB9_career   0.3666996  0.0268436  13.661   <2e-16 ***
velocity    -0.0219823  0.0085744  -2.564   0.0105 *
spin_rate   -0.0001996  0.0001322  -1.510   0.1315
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6185 on 821 degrees of freedom
Multiple R-squared:  0.3797,    Adjusted R-squared:  0.3767
F-statistic: 125.6 on 4 and 821 DF,  p-value: < 2.2e-16
```

The model is statistically significant (F-statistic - p-value < 2.2e-16) and explains approximately 37.7% of the variance in ERA_career (Adjusted R-squared = 0.3767). For each additional strikeout per 9 innings, a pitcher's career ERA is predicted to decrease by 0.26, holding other factors constant. This is a highly significant predictor, confirming that dominant, high-strikeout pitchers are more effective at preventing runs. For each additional walk per 9 innings, a pitcher's career ERA is predicted to increase by 0.37. This highly significant result underscores that poor control is hugely detrimental to a pitcher's success. It also permitted us to determine that for each 1 mph increase in average velocity, ERA is predicted to decrease by 0.022. This finding is statistically significant (p = 0.0105) and supports the idea that higher velocity contributes to better run prevention. Ultimately, the spin rate was not deemed a statistically significant predictor of ERA (p = 0.1315) when controlling the other variables.

Model 2: Predicting Total Innings Pitched (Durability)

The second model (lm_ip) was designed to predict a pitcher's total career innings pitched (IP_total), serving as a proxy for durability and long-term workload capacity (see Figure 5).

Figure 5

```
Call:
lm(formula = IP_total ~ SO9_career + BB9_career + velocity +
    spin_rate, data = pitcher_df)

Residuals:
    Min      1Q  Median      3Q     Max
-490.44 -184.76  -77.19  113.26 1230.97

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  530.85068  339.52150   1.564 0.118314
SO9_career    24.02756    6.92759   3.468 0.000551 ***
BB9_career  -141.73000   12.31261 -11.511  < 2e-16 ***
velocity       0.75902    3.93291   0.193 0.847013
spin_rate      0.03586    0.06065   0.591 0.554522
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 283.7 on 821 degrees of freedom
Multiple R-squared:  0.1428,    Adjusted R-squared:  0.1386
F-statistic: 34.19 on 4 and 821 DF,  p-value: < 2.2e-16
```

While the overall model is statistically significant (p-value < 2.2e-16), it explains only 13.9% of the variance in IP_total (Adjusted R-squared = 0.1386). This suggests that these specific metrics are not strong predictors of a pitcher's total career workload on their own. Pitchers with higher strikeout rates tend to accumulate more innings over their careers, likely because their effectiveness keeps them in pitching rotations, permitting us to identify it as a significant positive predictor. During our model evaluation, we observed that BB9_career was the strongest predictor, exhibiting a significant negative relationship. This indicates that for every additional walk per 9 innings, a pitcher is expected to throw approximately 142 fewer career innings. This suggests that pitchers with poor control receive far fewer opportunities over the long term. Lastly, regarding velocity and spin rate, neither of these "stuff" metrics was found to be a statistically significant predictor of the total innings a pitcher throws.

Splitting the Data

The dataset was then split into a training and a test set. The training dataset consists of the data on which the model will be trained, and the test set will be used to evaluate the model. This helps ensure that the model performs effectively on new data and that we're not overfitting the model. Therefore, 70% of the observations are used for the training set, and 30% are used for testing, ensuring the injury occurred response variable remains balanced across both sets. The training set comprised 578 observations with 48 variables, while the test set contained 248 observations with 48 variables. We're only going to be reporting the results of the confusion matrix and model performance of the logistic regression model based on the test set in this report, as the test set ensures that the model's predictive accuracy is assessed on unseen data, offering a more reliable measure of its generalizability. This approach also helps to avoid overfitting, where a model performs well on training data but poorly on new observations.

Logistic Regression Models

Two logistic regression models were fitted to the training data. The first logistic regression model included the variables: total_pitches, velocity, spin_rate, IP_total, SO9_career, and BB9_career. However, diagnostic evaluation revealed substantial multicollinearity between predictors total_pitches and IP_total, as they showed high VIF values. Since both metrics capture pitcher workload, this indicates strong multicollinearity:

total_pitches	velocity	spin_rate	IP_total	SO9_career	BB9_career

42.278779	1.248559	1.149305	43.837395	1.294158	1.403822

To improve model stability and interpretability, only total_pitches were retained in the final model, since they offer a direct measure of cumulative workload and injury risk. That is why we used the final logistic regression model for analysis. It had the same variables, except that IP_total was removed. This logistic regression model predicts the probability that a pitcher experienced an injury using key metrics about their workload and performance.

Logistic Regression Model: Predicting Injury Occurrence

The logistic regression model was fit to predict injury occurrence among pitchers using total_pitches, velocity, spin_rate, SO9_career, and BB9_career as predictors (see Figure 6). Multicollinearity was also checked for this model:

total_pitches	velocity	spin_rate	SO9_career	BB9_career
1.121891	1.220109	1.143447	1.295314	1.238973

Since the VIF values are less than 5, this suggests that multicollinearity isn't a problem for this model. The independent variables in the regression model are not correlated.

Figure 6

```
Call:
glm(formula = injury_occurred ~ total_pitches + velocity + spin_rate +
    SO9_career + BB9_career, family = binomial, data = train)

Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)    5.350e+00  3.143e+00   1.702   0.0887 .
total_pitches  1.216e-04  2.117e-05   5.743 9.29e-09 ***
velocity      -6.184e-02  3.677e-02  -1.682   0.0926 .
spin_rate     -3.145e-04  5.346e-04  -0.588   0.5563
SO9_career    -8.324e-02  6.108e-02  -1.363   0.1729
BB9_career     1.322e-01  1.140e-01   1.159   0.2463
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 784.58  on 577  degrees of freedom
Residual deviance: 737.34  on 572  degrees of freedom
AIC: 749.34

Number of Fisher Scoring iterations: 4
```

The model revealed that total pitches were again highly statistically significant and had a lower p-value (9.29e-09) than the previous model. Velocity in this model showed a marginally negative association with injury occurrence. However, the rest of the predictors again were not statistically significant. The AIC (749.34) is a little higher in this model; however, even though the AIC was a little higher than the first model, this model improves interpretability and is more stable, as there is no multicollinearity present.

Since the regression coefficients are in log-odds, we decided to convert them to odds for interpretability. But before we did, these were our log-odd results:

(Intercept)	total_pitches	velocity	spin_rate	SO9_career	BB9_career
5.3498762786	0.0001215989	-0.0618442511	-0.0003145439	-0.0832426375	0.1321859697

and these were our odds results:

(Intercept)	total_pitches	velocity	spin_rate	SO9_career	BB9_career
210.5822427	1.0001216	0.9400293	0.9996855	0.9201279	1.1413206

The intercept is 210.5822427, meaning that when all the predictors were zero, the injury odds for a pitcher are 210.5822427. For total_pitches (1.0001216), it means that for each additional pitch, the odds of injury increase by a factor of 1.0001216, while holding other variables constant. For velocity (0.9400293), it means that for each additional 1 mph increase in average velocity, the odds of injury decrease by a factor of 0.9400293, while holding other variables constant. Though this is marginally significant, it may have a slight effect on injury odds. For spin_rate, SO9_career, and BB9_career, these essentially were not significant, so they didn't predict a pitcher's injury odds. It seems that total_pitches were the strongest and most reliable predictor of injury, which means workload was the dominant predictor of injury risk.

To further illustrate the model's predictions, probabilities were calculated by varying one predictor while holding others at their mean value. In this case, since total_pitches is highly statistically significant, probability changes for different values of total pitches were observed (see Figure 7).

Figure 7

	total_pitches	velocity	spin_rate	SO9_career	BB9_career	probs
1	2001.000	88.89528	2223.019	8.549378	3.31555	0.2937776
2	6265.996	88.89528	2223.019	8.549378	3.31555	0.4113256
3	25820.000	88.89528	2223.019	8.549378	3.31555	0.8828041

As the value of total_pitches increased from its minimum (2001) to maximum (25820), the probability of a pitcher getting injured increased from (0.2937776) to (0.8828041). These results clearly show that cumulative workload is strongly associated with a higher risk of injury, even when considering other performance-related metrics such as velocity, spin rate, strikeout rate, and walk rate.

Confusion Matrix and Model Performance: Test Set

As previously mentioned, the logistic regression model was assessed on the test set to evaluate predictive performance on unseen data (see Figure 8). The confusion matrix on the test set revealed that there were 138 true negatives (TN), so predicted no injury, and the pitcher did not get injured, 57 were false negatives (FN), so predicted no injury but the pitcher did get injured, 18 were false positives (FP), so predicted injury but the pitcher did not get injured, and 35 were true positives (TP), so predicted injury and the pitcher did get injured. On the test set, the model achieved an accuracy of 70%, so the model correctly predicted injury status 70% of the time, slightly higher than the 63% observed in the training data. A precision of 66%, so when the model predicted an injury, it was right about 66% of the time. A recall of 38%, suggesting that the model correctly identified roughly 38% of pitchers who sustained injuries. Specificity was higher at about 88%, indicating that the model was better at correctly identifying healthy players. While the model demonstrates moderate accuracy, its ability to identify injured players could be improved. Nonetheless, the significant contribution of total pitches as a predictor supports the conclusion that workload significantly increases injury risk for pitchers.

Figure 8

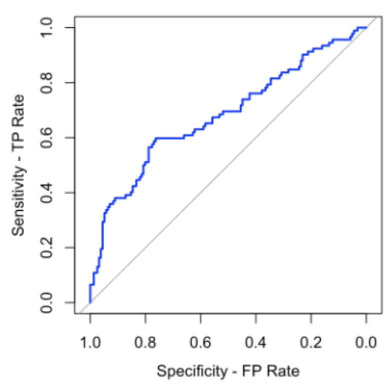
```
Confusion Matrix and Statistics

          Reference
Prediction  No Yes
       No  138  57
       Yes  18  35

               Accuracy : 0.6976
                 95% CI : (0.6363, 0.7541)
    No Information Rate : 0.629
    P-Value [Acc > NIR] : 0.01413

                  Kappa : 0.2903

 Mcnemar's Test P-Value : 1.145e-05

            Sensitivity : 0.3804
            Specificity : 0.8846
         Pos Pred Value : 0.6604
         Neg Pred Value : 0.7077
             Prevalence : 0.3710
         Detection Rate : 0.1411
   Detection Prevalence : 0.2137
      Balanced Accuracy : 0.6325

       'Positive' Class : Yes
```

The results compared to the training set are fairly consistent, indicating that the model generalizes well to unseen data (test set) and is not overfit. The results, as shown in the Draft Report for the confusion matrix and model performance on the training set, were: 283 (TN), 158 (FN), 55 (FP), and 82 (TP). An accuracy of 63%, a precision of about 60%, a recall of 34%, and a specificity of about 84%. However, it underperforms while using the train set as the model might be learning the expected outcomes instead of properly running the statistical analysis. These findings again illustrate that the model captures meaningful relationships between workload and the probability of injury; however, improvements in identifying injury cases could be achieved.

ROC Curve and AUC

A ROC Curve was then created for the test set to illustrate the trade-off between sensitivity (true positive rate) and specificity (false positive rate) across different classification thresholds (see Figure 9). The ROC curve line lies above the diagonal reference line, suggesting a separation and distinction between injured and non-injured pitchers. Even though it doesn't do it perfectly, logistic regression can meaningfully rank pitchers by injury risk, and it does it better than random guessing. The area under the curve (AUC) was 0.682, suggesting that the model has fair discriminatory power. This indicates that the logistic regression model can correctly distinguish between injured and non-injured pitchers by approximately 68% of the time. Although the model is not highly discriminative, this performance level is reasonable considering the complex and multifactorial nature of sports injuries.

Figure 9



IL TREND

To enhance the context of the logistic regression analysis, a summary of the pitchers who appeared most frequently on the Injured List (IL) was created using the injury_count variable (see Figure 10). This variable represented the total number of times each pitcher was recorded on the IL between 2015 and 2017. Ranking and displaying the top ten pitchers by their IL appearances revealed that if an injury occurs more than once, then they're more prone to appearing in the IL again. It is also known that pitchers undergoing a Tommy John Surgery due to a UCL tear can spend a full season on the IL list, indeed creating further impact on this overall analysis, as it is important to not only consider the times a player gets injured but also the days he spends in the injured list.

When we further examine this group's statistics during their injury years, we can observe that pitching performance metrics decreased. There's also a notable increase in these injuries as pitchers age and sustain excessive workloads, leading to injury placement during these years due to the total pitches thrown, which produces additional muscular stress and results in increased injury development.

Figure 10

	player_name	injury_count
1	McCarthy, Brandon	6
2	Cain, Matt	5
3	Garza, Matt	5
4	Griffin, A.J.	5
5	Hill, Rich	5
6	Petricka, Jake	5
7	Ryu, Hyun Jin	5
8	Sanchez, Aaron	5
9	Strasburg, Stephen	5
10	Wheeler, Zack	5

Conclusion

Key Takeaways & Strategic Recommendations

Conclusively, the information obtained through the effective development of the logistic regression analysis upon the incorporation of IL placement was enticing as it permitted us to clearly visualize those factors truly resulting into an adverse effect towards pitcher's health and the importance of managers, coaches and team executives in having a proper risk management plan in place to prevent the drastic amount of injuries from occurring. This would be done using procedures that provide data-driven workload monitoring techniques to support predictive metrics to anticipate fatigue before injuries occur, and combining it with preventive strategies that incorporate biomechanical evaluations, load management, and off-days optimization for high-pitch-volume pitcher profiles, usually comprised of well-established pitchers and coveted pitching prospects.

It is known that 2025 has been a year marked with injuries due to the increase in "max-effort"; this is something that goes beyond the scope of the data being analyzed, but definitely correlates with the unresolved results demonstrating that pitching workload is the most crucial factor that eventually all organizations must take into account at the time of elaborating detailed scheduling structures that prioritizes pitcher's health to stretch their overall longevity as the data demonstrated that velocity and workload (total pitches) are key predictors of injury risk as higher pitch counts correlate with increased odds of IL placement, while moderate velocity may reduce injury odds. Indeed, in the real world, you not only need to minimize the amount of BB9 but also keep the SO9 at a reasonable rate without affecting competition levels, understanding that there are other means of getting batters out.

Limitations & Future Research

Understanding that the data had some constraints due to the information publicly available for injured baseball players we could improve the impactful insights derived from this report by matching the IL data to the years included from Savant Statcast level data and Lahman Pitching dataset. It was important to delimit the data to at least 2,000 pitches thrown to avoid unnecessary noise produced by pitchers who had a small stint during these 9 seasons and position players who had limited appearances. If we had divided the data into starting and relief pitchers, we would also have been able to enhance the quality of our overall analytical research paper, thoroughly assessing the depth of the analysis and understanding the differences showcased between pitchers in diverse roles. Incorporating fatigue metrics, recovery days and age as new predictors will continue to add value to the already reached conclusions and through use of advanced ML models for nonlinear injury prediction, we would elevate the strength of it to the next level!

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